

Maximum Xor Subset

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, your task is to find the maximum xor sum of a subset.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the maximum xor sum of a subset.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^9$

Example

Input	Output
4 1 6 12 6	13